

Simple Alarm Clock

🎯 AIM

“Develop a Simple Alarm Clock application in Python that enables users to set an alarm for a specific time. The program should continuously monitor the system's current time using Python's built-in date and time modules. Once the current system time matches the user-specified alarm time, the program should notify the user by displaying an alert message and optionally playing a sound.”

STUDENT DETAILS

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SECTION

Section B

BATCH INCHARGE

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Levels of My Project

MODULAR STRUCTURE

1

Level 1 – User Input

AIM

Allow the user to enter and set the desired alarm time.

2

Level 2 – Time Monitoring

AIM

Continuously obtain and monitor the current system time using Python's built-in date and time modules.

3

Level 3 – Time Comparison

AIM

Compare the current system time with the user-specified alarm time.

4

Level 4 – Alarm Notification

AIM

When the current time matches the specified alarm time, display an alert message and optionally play a sound.

Execution Flow: Level1 → Level2 → Level3 → Level4

Introduction and Problem Statement

PROJECT OVERVIEW

Introduction

- ▶ A Simple Alarm Clock is a Python application that allows users to set an alarm for a particular time.
- ▶ The application continuously checks the current system time.
- ▶ Python's built-in date and time modules are used to obtain the current time.
- ▶ The current time is compared with the alarm time entered by the user.
- ▶ When both times match, the application gives an alert to the user.
- ▶ A sound can optionally be played along with the alert.

Problem Statement

“Develop a Simple Alarm Clock application in Python that enables users to set an alarm for a specific time. The program should continuously monitor the system's current time using Python's built-in date and time modules. Once the current system time matches the user-specified alarm time, the program should notify the user by displaying an alert message and optionally playing a sound.”

What I Made in Module 1 & Concepts Used

IMPLEMENTATION DETAILS

Work done in Module 1

- ✓ Takes the desired alarm time from the user as input.
- ✓ Continuously fetches the exact current system time.
- ✓ Formats and compares the current time with the alarm time perfectly.
- ✓ Uses a continuous loop to monitor time without closing.
- ✓ Triggers and prints an alert message when the target time is reached.

Python concepts used



Variables

To store alarm string and current time data



datetime module

To get exact current system time automatically



while loop

To repeat time-checking process continuously



if statement

To compare current time with user's alarm time

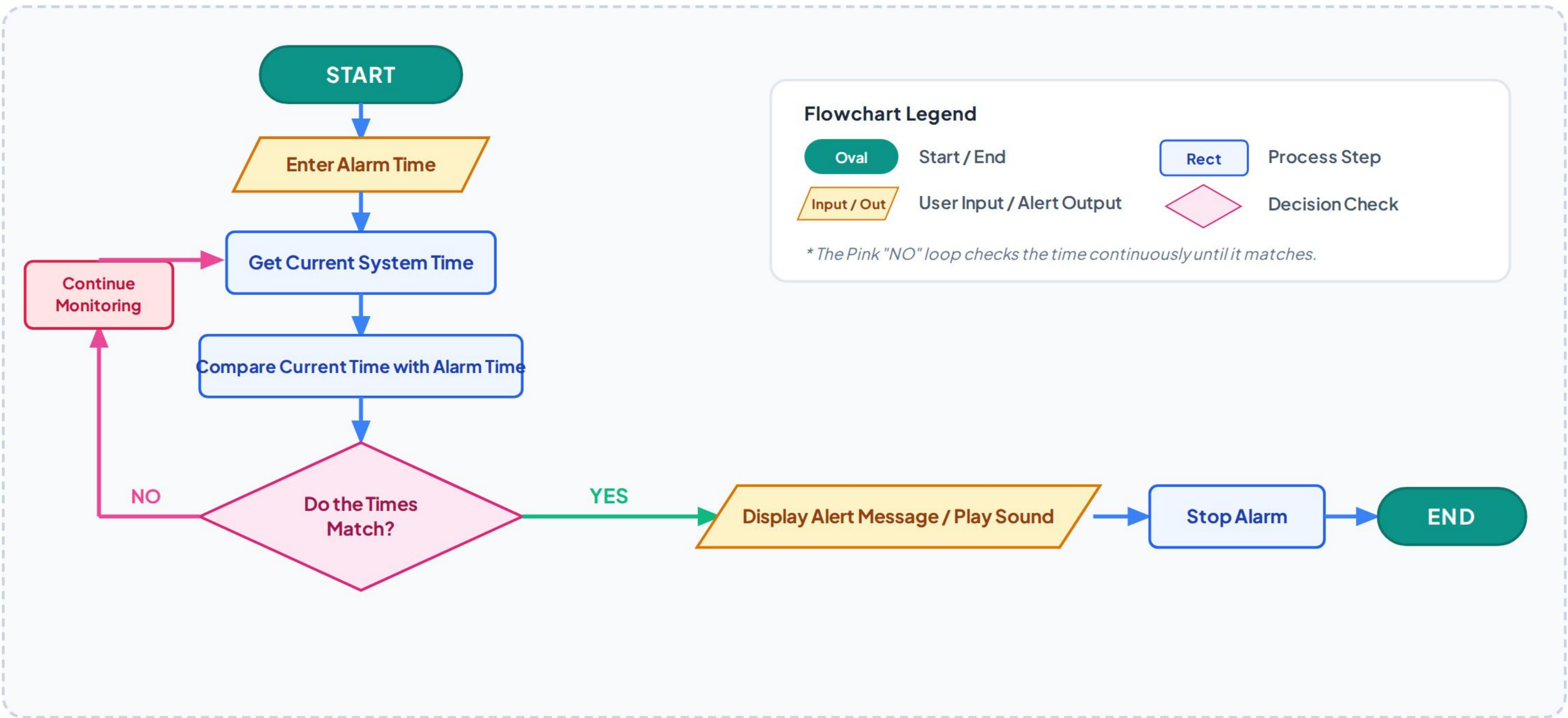


break statement

To stop the loop once the alarm rings

Flowchart of Simple Alarm Clock

EXECUTION LOGIC



Conclusion

SUMMARY

- ✓ The Simple Alarm Clock application allows users to set an alarm for a specific time easily.
- ✓ The application continuously compares the current time with the user-specified alarm target.
- ✓ An alarm sound can optionally be played to physically notify and wake the user.
- ✓ Python's built-in date and time modules efficiently monitor the exact current system time.
- ✓ When the current time matches the alarm time, a printed alert message is instantly displayed.
- ✓ The project successfully demonstrates core Python concepts like inputs, loops, and condition checking.

THANK YOU